

underlying molecular mechanisms that regulate mitochondrial biogenesis, the process by which new mitochondria are formed. These efforts resulted in the discovery of *peroxisome proliferator-activated receptor-γ coactivator-1α* (PGC-1α), a key regulator protein that is integrally involved in mitochondrial biogenesis in skeletal muscle. Because of its multiple important roles in enhancing metabolic function, PGC-1α is often called the master regulator or master switch. It is also now well established that both acute exercise and exercise training—both endurance and resistance exercise—enhance PGC-1α expression.

As shown in [figure 11.9](#), exercise training promotes biogenesis of new mitochondria, slows the decline in mitochondrial function by remodeling mitochondria through processes of fusion and fission, and helps maintain mitophagy in skeletal muscle. Thus, mitochondrial quality control is an important exercise-induced adaptation.³⁶

Increased expression of PGC-1α protein can be measured in skeletal muscle even after a single bout of exercise; after two or three repeated bouts, markers for mitochondrial biogenesis can be observed. Increased PGC-1α not only increases mitochondrial biogenesis but also controls the replacement of old weakened mitochondria with new healthy mitochondria. Mitochondrial damage induced by such insults as hypoxia, inflammation, or increased oxidant stress can lead to the accumulation of metabolic by-products that impair mitochondrial function. Although addition of new mitochondria is of extreme importance, the maintenance of a healthy population of mitochondria is equally critical for optimal metabolic capacity. Continuous removal of damaged mitochondria is likewise important for optimal function of skeletal muscle.

Oxidative Enzymes

Regular endurance exercise has been shown to induce major adaptations in skeletal muscle, including an increase in the number and size of the muscle fiber mitochondria as just discussed. These changes are further enhanced by an increase in mitochondrial capacity. The oxidative breakdown of fuels and the ultimate production of ATP depend on the action of **mitochondrial oxidative enzymes**, the specialized proteins that catalyze (i.e., speed up) the

breakdown of nutrients to form ATP. Aerobic training increases the activity of these important enzymes.

Figure 11.10 illustrates the changes in the activity of succinate dehydrogenase (SDH), a key muscle oxidative enzyme, over 7 months of progressive swim training. While the rate of increases in $\dot{V}O_{2\max}$ slowed after the first 2 months of training, activity of this key oxidative enzyme continued to increase throughout the entire training period. This suggests that training-induced increases in $\dot{V}O_{2\max}$ might be limited more by the circulatory system's ability to transport oxygen than by the muscles' oxidative potential.

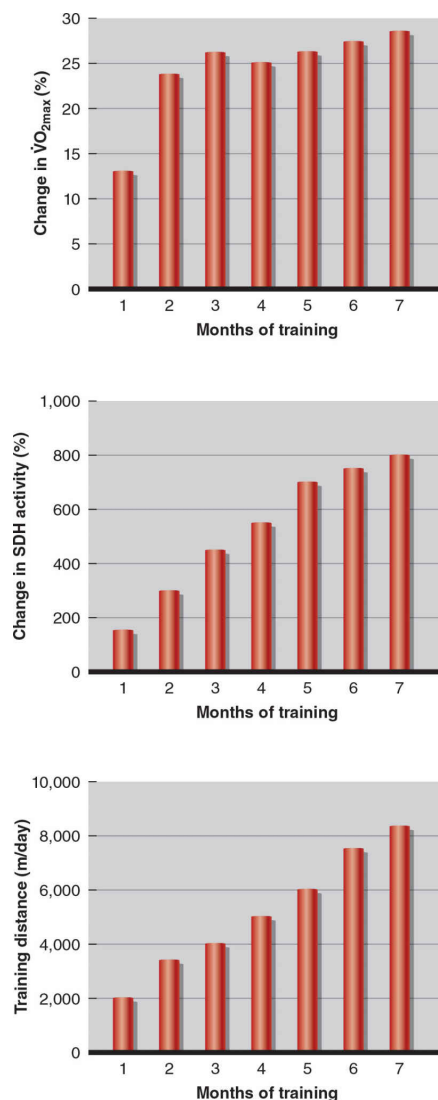
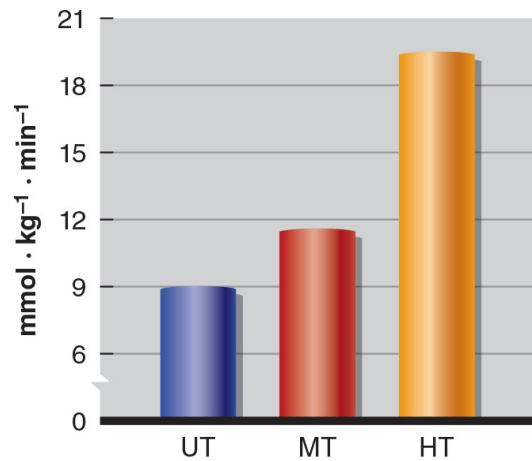


FIGURE 11.10 The percentage change in maximal oxygen uptake ($\dot{V}O_{2\max}$) and the activity of succinate dehydrogenase (SDH), one of the muscles' key oxidative enzymes, during 7 months of swim training. Interestingly, although this enzyme activity continues to increase with increasing levels of

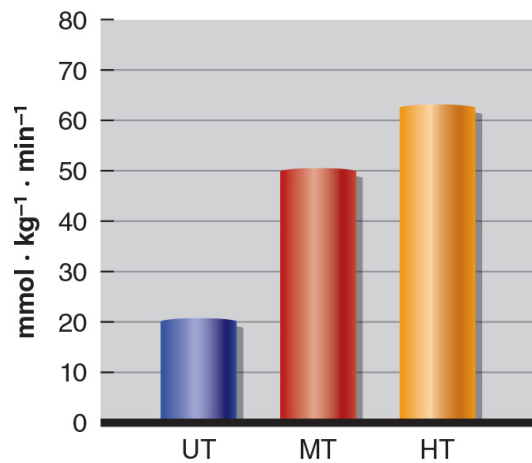
training, the swimmers' maximal oxygen uptake appears to level off after the first 8 to 10 weeks of training. This implies that mitochondrial enzyme activity is not a direct indication of whole-body endurance capacity.

The activities of muscle enzymes such as SDH and citrate synthase are dramatically influenced by aerobic training. This is seen in [figure 11.11](#), which compares the activities of these enzymes in untrained people, moderately trained joggers, and highly trained runners.⁹ Even moderate daily exercise increases the activity of these enzymes and thus the oxidative capacity of the muscle. For example, jogging or cycling for as little as 20 min per day has been shown to increase SDH activity in leg muscles by more than 25%. Training more vigorously—for example, for 60 to 90 min per day—produces a two- to threefold increase in this enzyme's activity.

One metabolic consequence of mitochondrial changes induced by aerobic training is **glycogen sparing**, a slower rate of utilization of muscle glycogen and enhanced reliance on fat as a fuel source at a given exercise intensity. Enhanced glycogen sparing with endurance training most likely improves the ability to sustain a higher exercise intensity, such as maintaining a faster race pace in a 10 km run.



a Succinate dehydrogenase



b Citrate synthase

FIGURE 11.11 Leg muscle (gastrocnemius) enzyme activities of untrained (UT) subjects, moderately trained (MT) joggers, and highly trained (HT) marathon runners. Enzyme levels are shown for two of many key enzymes that participate in the oxidative production of adenosine triphosphate.

Adapted from Costill, Fink, Lesmes, et al. (1979); Costill, Coyle, Fink, et al. (1979).

In summary, endurance exercise training causes a wide variety of phenotypic adaptations in skeletal muscle, including angiogenesis (creation of new capillaries), transformation of fiber types from glycolytic to oxidative, increased ability to mobilize and use fats as a substrate, and increased glucose uptake by muscle fibers, which increases the number of mitochondria and improves the overall quality of the existing mitochondrial pool.

In Review

- Aerobic training selectively recruits type I muscle fibers and fewer type II fibers. Consequently, the type I fibers increase their cross-sectional area with aerobic training.
 - After training, there appears to be a small increase in the percentage of type I fibers, as well as a transition of some type IIx to type IIa fibers.
 - Aerobic training increases both the number of capillaries per muscle fiber and the number of capillaries for a given cross-sectional area of muscle. These changes improve blood perfusion through the muscles, enhancing the diffusion of oxygen, carbon dioxide, nutrients, and by-products of metabolism between the blood and muscle fibers.
 - Aerobic training increases muscle myoglobin content by as much as 80%. Myoglobin transports oxygen from cell membranes to the mitochondria.
 - Aerobic training increases both the number and the size of muscle fiber mitochondria, providing the muscle with an increased capacity for oxidative metabolism.
 - Endurance exercise training also improves the overall quality of the existing mitochondrial pool.
 - Activities of many oxidative enzymes are increased with aerobic training.
 - These changes occurring in the muscles, combined with adaptations in the oxygen transport system, enhance the capacity of oxidative metabolism and improve endurance performance.
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Metabolic Adaptations to Training

Now that we have discussed training changes in both the cardiovascular and respiratory systems, as well as skeletal muscle adaptations, we are ready to examine how these integrated adaptations are reflected by changes in three important physiological variables related to metabolism:

- Lactate threshold
- Respiratory exchange ratio
- Oxygen consumption

Lactate Threshold

Lactate threshold, discussed in [chapter 5](#), is a physiological marker that is closely associated with endurance performance—the higher the lactate threshold, the better the performance capacity. [Figure](#)

[11.12a](#) illustrates the difference in lactate threshold between an endurance-trained individual and an untrained individual. This figure also accurately represents the changes in lactate threshold that would occur following a 6- to 12-month program of endurance training. In either case, in the trained state, one can exercise at a higher percentage of one's $\dot{V}O_{2\max}$ before lactate begins to accumulate in the blood. In this example, the trained runner could sustain a race pace of 70% to 75% of $\dot{V}O_{2\max}$, an intensity that would result in continued lactate accumulation in the blood of the untrained runner. This translates into a much faster race pace (see [figure 11.12b](#)). Above the lactate threshold, the lower lactate at a given rate of work is likely attributable to a combination of reduced lactate production and increased lactate clearance. As athletes become better trained, their postexercise blood lactate concentrations are lower for a given rate of work.

Respiratory Exchange Ratio

Recall from [chapter 5](#) that the respiratory exchange ratio (RER) is the ratio of carbon dioxide released to oxygen consumed during metabolism. The RER reflects the composition of the mixture of substrates being used as an energy source, with a lower RER reflecting an increased reliance on fats for energy production and a higher RER reflecting a higher contribution of carbohydrates.

After training, the RER decreases at both absolute and relative submaximal exercise intensities. These changes are attributable to a greater utilization of free fatty acids instead of carbohydrate at these work rates following training.

Resting and Submaximal Oxygen Consumption

Oxygen consumption ($\dot{V}O_2$) at rest is unchanged following endurance training. While a few cross-sectional comparisons have suggested that training elevates resting $\dot{V}O_2$, the HERITAGE Family Study—with a large number of subjects and with duplicate measures of resting metabolic rate both before and after 20 weeks of training—showed no evidence of an increased resting metabolic rate after training.³⁵

During submaximal exercise at a given intensity, $\dot{V}O_2$ is either unchanged or slightly reduced following training. In the HERITAGE

Family Study, training reduced submaximal $\dot{V}O_2$ by 3.5% at a work rate of 50 W. There was a corresponding reduction in cardiac output at 50 W, reinforcing the strong interrelationship between $\dot{V}O_2$ and cardiac output.³⁴ This small decrease in $\dot{V}O_2$ during submaximal exercise, not seen in many studies, could have resulted from an increase in exercise economy (performing the same exercise intensity with less extraneous movement).

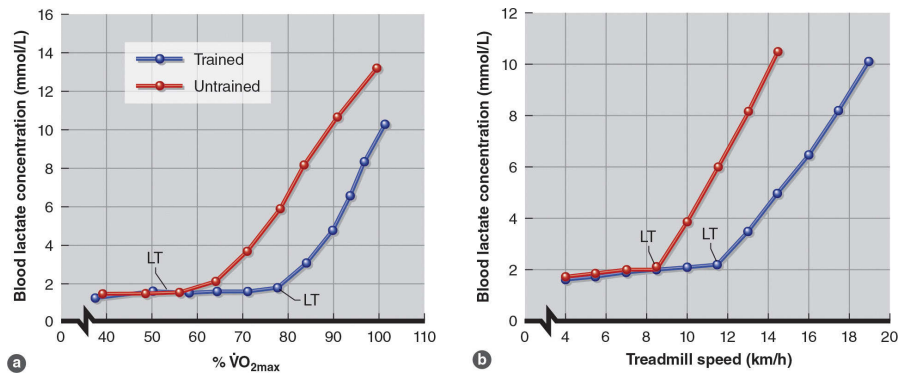


FIGURE 11.12 Changes in lactate threshold (LT) with training expressed as (a) a percentage of maximal oxygen uptake ($\% \dot{V}O_{2max}$) and (b) an increase in speed on the treadmill. Lactate threshold occurs at a speed of 8.4 km/h (5.2 mph) in the untrained state and at 11.6 km/h (7.2 mph) in the trained state.

Maximal Oxygen Consumption

$\dot{V}O_{2max}$ is the best indicator of cardiorespiratory endurance capacity and increases substantially in response to endurance training. While small and very large increases have been reported, an increase of 15% to 20% is typical for a previously sedentary person who trains at 50% to 85% of his or her $\dot{V}O_{2max}$ three to five times per week, 20 to 60 min per day, for 6 months. For example, the $\dot{V}O_{2max}$ of a sedentary individual could reasonably increase from $35 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ to $42 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ as a result of such a program. This is far below the values we see in world-class endurance athletes, whose values generally range from 70 to $94 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. The more sedentary an individual is when starting an exercise program, the larger the increase in $\dot{V}O_{2max}$.

Integrated Adaptations to Chronic Endurance Exercise

It should now be clear that the adaptations that accompany endurance training are many and that they affect multiple physiological systems. Physiologists commonly establish models to help explain how various physiological factors or variables work together to affect a specific outcome or component of performance. Dr. Donna H. Korzick, an exercise physiologist at Pennsylvania State University, has created a unifying figure to model the factors that contribute to the cardiovascular adaptation to chronic endurance training (see figure 11.13).

What Limits Aerobic Power and Endurance Performance?

A number of years ago, exercise scientists were divided on what major physiological factor or factors actually limit $\dot{V}O_{2\max}$. Two contrasting theories had been proposed.

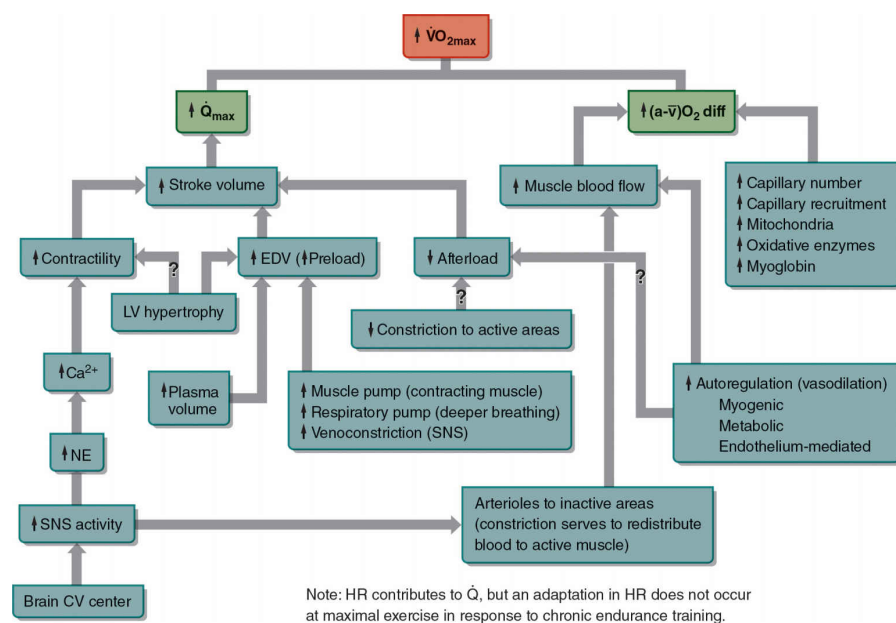


FIGURE 11.13 Cardiovascular adaptations to chronic endurance exercise.

Adapted by permission from Donna H. Korzick, Pennsylvania State University, 2006.

One theory held that endurance performance was limited by the lack of sufficient concentrations of oxidative enzymes in the mitochondria. Endurance training programs substantially increase these oxidative enzymes, allowing active tissue to use more of the available oxygen, resulting in a higher $\dot{V}O_{2\max}$. In addition, endurance training increases both the size and number of muscle mitochondria.

Thus, this theory argued, the main limitation of maximal oxygen consumption is the inability of the existing mitochondria to use the available oxygen beyond a certain rate. This theory was referred to as the utilization theory.

The second theory proposed that central and peripheral cardiovascular factors limit endurance capacity. These circulatory influences would preclude delivery of sufficient amounts of oxygen to the active tissues. Taking into account the observation that improvement in $\dot{V}O_{2\max}$ following endurance training results from increased blood volume, increased cardiac output (via stroke volume), and a better perfusion of active muscle with blood, this theory proposed that these cardiovascular factors are the limiting factor for $\dot{V}O_{2\max}$.

Evidence strongly supports the latter theory. In one study, subjects breathed a mixture of carbon monoxide (which irreversibly binds to hemoglobin, limiting hemoglobin's oxygen-carrying capacity) and air during exercise to exhaustion.²⁶ $\dot{V}O_{2\max}$ decreased in direct proportion to the percentage of carbon monoxide breathed. The carbon monoxide molecules bonded to approximately 15% of the total hemoglobin; this percentage agreed with the percentage reduction in $\dot{V}O_{2\max}$. In another study, approximately 15% to 20% of each subject's total blood volume was removed.¹¹ $\dot{V}O_{2\max}$ decreased by approximately the same relative amount. Reinfusion of the subjects' packed red blood cells approximately 4 weeks later increased $\dot{V}O_{2\max}$ well above baseline or control conditions. In both studies, the reduction in the oxygen-carrying capacity of the blood—via either blocking hemoglobin or removing whole blood—resulted in the delivery of less oxygen to the active tissues and a corresponding reduction in $\dot{V}O_{2\max}$. Similarly, studies have shown that breathing oxygen-enriched mixtures, in which the partial pressure of oxygen in the inspired air is substantially increased, increases endurance capacity.

These and subsequent studies indicated that the available oxygen supply is the major limiter of endurance performance. Oxygen transport to the working muscles, not the available mitochondria and oxidative enzymes, limits $\dot{V}O_{2\max}$. The argument was that increases in

$\dot{V}O_{2\max}$ with training are largely attributable to increased maximal blood flow and increased muscle capillary density in the active tissues. Skeletal muscle adaptations (including increased mitochondrial content and respiratory capacity of the muscle fibers) contribute importantly to the ability to perform prolonged, high-intensity, submaximal exercise.

Table 11.3 summarizes the typical physiological changes that occur with endurance training. The values (pre- and posttraining) for a previously inactive man are compared with values for a world-class male endurance runner.

In Review

- Lactate threshold increases with endurance training, allowing performance of higher exercise intensities without significantly increasing blood lactate concentration.
- With endurance training, the RER decreases at submaximal work rates, indicating greater utilization of free fatty acids as an energy substrate (carbohydrate sparing).
- Oxygen consumption generally remains unchanged at rest and remains unaltered or decreases slightly during submaximal exercise following endurance training.
- $\dot{V}O_{2\max}$ increases substantially following endurance training, but the extent of increase possible is genetically limited in each individual. The major limiting factor appears to be oxygen delivery to the active muscles.

Long-Term Improvement in Aerobic Power and Cardiorespiratory Endurance

Although an individual's highest attainable $\dot{V}O_{2\max}$ is usually achieved within 12 to 18 months of intense endurance training, endurance *performance* can continue to improve. Improvement in endurance performance without improvement in $\dot{V}O_{2\max}$ is likely attributable to improvements in the ability to perform at increasingly higher percentages of $\dot{V}O_{2\max}$ for extended periods. Consider, for example, a young male runner who starts training with an initial $\dot{V}O_{2\max}$ of $52.0 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. He reaches his genetically determined peak $\dot{V}O_{2\max}$ of $71.0 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ after 2 years of intense training, after which no further increases occur, even with more frequent or more intense

workouts. At this point, as shown in [figure 11.14](#), the young runner is able to run at 75% of his $\dot{V}O_{2\max}$ ($0.75 \times 71.0 = 53.3 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) in a 10 km (6.2 mi) race. After an additional 2 years of intensive training, his $\dot{V}O_{2\max}$ is unchanged, but he is now able to compete at 88% of his $\dot{V}O_{2\max}$ ($0.88 \times 71.0 = 62.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$). Obviously, by being able to sustain an oxygen uptake of $62.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, he is able to run at a much faster race pace.

This ability to sustain exercise at a higher percentage of $\dot{V}O_{2\max}$ is partly the result of an increase in the ability to buffer lactate, because race pace is directly related to the $\dot{V}O_2$ value at which lactate begins to accumulate.

TABLE 11.3 Typical Effects of Endurance Training in a Previously Inactive Man, Contrasted with Values for a Male World-Class Endurance Athlete

Variables	Pretraining, sedentary male	Posttraining, sedentary male	World-class endurance athlete
Cardiovascular			
HR _{rest} (beats/min)	75	65	45
HR _{max} (beats/min)	185	183	174
SV _{rest} (ml/beat)	60	70	100
SV _{max} (ml/beat)	120	140	200
$\dot{Q}$ at rest (L/min)	4.5	4.5	4.5
$\dot{Q}_{\max}$ (L/min)	22.2	25.6	34.8
Heart volume (ml)	750	820	1,200
Blood volume (L)	4.7	5.1	6.0
Systolic BP at rest (mmHg)	135	130	120
Systolic BP _{max} (mmHg)	200	210	220
Diastolic BP at rest (mmHg)	78	76	65
Diastolic BP _{max} (mmHg)	82	80	65
Respiratory			
$\dot{V}_E$ at rest (L/min)	7	6	6
$\dot{V}_E$ max (L/min)	110	135	195
TV at rest (L)	0.5	0.5	0.5
TV _{max} (L)	2.75	3.00	3.90
VC (L)	5.8	6.0	6.2
RV (L)	1.4	1.2	1.2
Metabolic			
(a- $\bar{V}O_2$ diff at rest (ml/100 ml)	6.0	6.0	6.0
(a- $\bar{V}O_2$ diff max (ml/100 ml)	14.5	15.0	16.0
$\dot{V}O_2$ at rest ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	3.5	3.5	3.5
$\dot{V}O_{2\max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	40.7	49.9	81.9
Blood lactate at rest (mmol/L)	1	1	1
Blood lactate max (mmol/L)	7.5	8.5	9.0

Body composition			
Weight (kg)	79	77	68
Fat weight (kg)	12.6	9.6	5.1
Fat-free weight (kg)	66.4	67.4	62.9
Fat (%)	16.0	12.5	7.5

Note. HR = heart rate; SV = stroke volume; $\dot{Q}$ = cardiac output; BP = blood pressure; $\dot{V}_E$ = ventilation; TV = tidal volume; VC = vital capacity; RV = residual volume; (a- $\bar{V}$ O₂ diff = arterial-mixed venous oxygen difference; $\dot{V}$ O₂ = oxygen consumption.

Factors Affecting an Individual's Response to Aerobic Training

We have discussed general trends in adaptations that occur in response to endurance training. However, we must always remember that we are talking about adaptations in individuals and that everyone does not respond in the same manner. Several factors that can affect individual response to aerobic training must be considered.

Training Status and $\dot{V}O_{2max}$

The higher the initial state of conditioning, the smaller the relative improvement for the same volume of training. For example, if two people, one sedentary and the other partially trained, undergo the same endurance training program, the sedentary person will show the greatest relative (%) improvement.

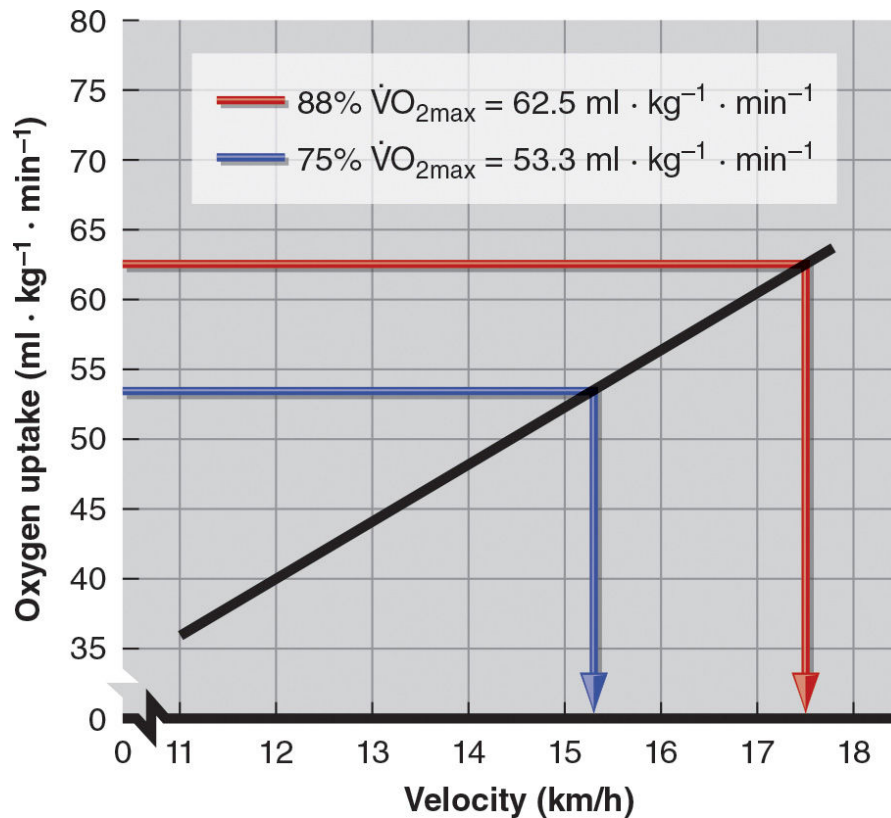


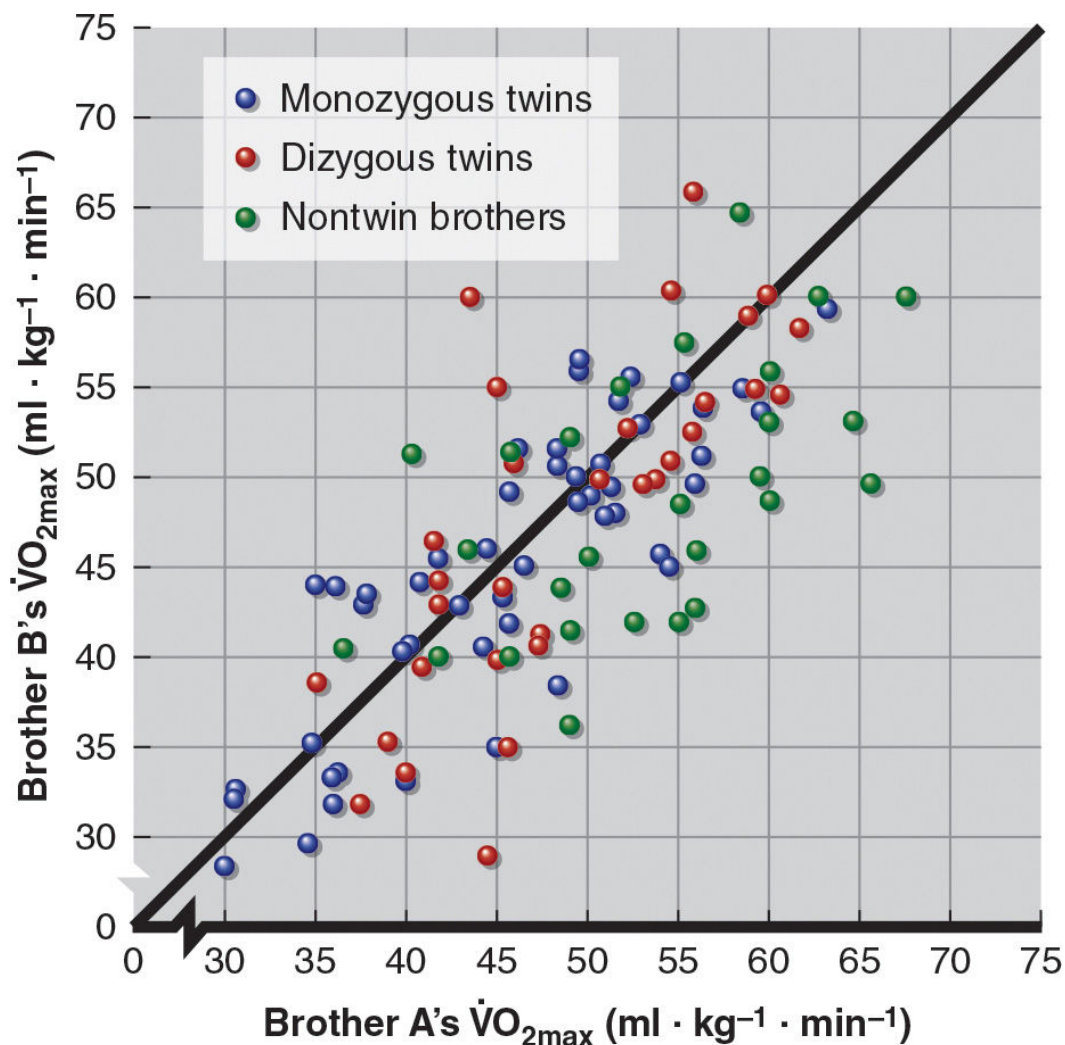
FIGURE 11.14 Change in race pace with continued training after maximal oxygen uptake stops increasing beyond $71 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$.

In fully mature athletes, the highest attainable $\dot{V}O_{2max}$ is reached within 8 to 18 months of intense endurance training, indicating that each athlete has a finite maximal attainable level of oxygen consumption. This finite range is genetically determined but may potentially be influenced by training in early childhood during the development of the cardiovascular system.

Heredity

The ability to increase maximal oxygen consumption levels is genetically limited. This does not mean that each individual has a preprogrammed $\dot{V}O_{2max}$ that cannot be exceeded. Rather, a range of $\dot{V}O_{2max}$ values seems to be predetermined by an individual's genetic makeup, with that individual's highest attainable $\dot{V}O_{2max}$ somewhere in that range. Each individual is born into a predetermined genetic window, and the person can shift up or down within that window with exercise training or detraining, respectively.

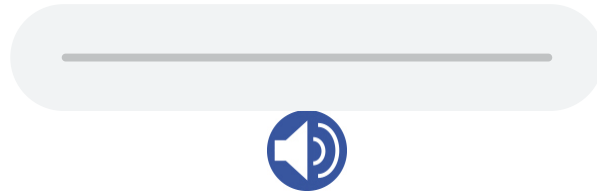
Research on the genetic basis of $\dot{V}O_{2\max}$ began in the late 1960s and early 1970s. Recent research has shown that identical (monozygous) twins have similar $\dot{V}O_{2\max}$ values, whereas the variability for dizygous (fraternal) twins is much greater (see [figure 11.15](#)).⁵ Each symbol represents a pair of brothers. Brother A's $\dot{V}O_{2\max}$ value is indicated by the symbol's position on the x-axis, and brother B's $\dot{V}O_{2\max}$ value is on the y-axis. Similarity in the siblings' $\dot{V}O_{2\max}$ values is noted by comparing the x and y coordinates of the symbol (i.e., how close it falls to the diagonal line $x = y$ on the graph). Similar results were found for endurance capacity, determined by the maximal amount of work performed in an all-out, 90 min ride on a cycle ergometer.



Adapted by permission from Bouchard et al. (1986).

FIGURE 11.15 Comparisons of $\dot{V}O_{2\max}$ in twin (monozygous and dizygous) and nontwin brothers.

Adapted by permission from C. Bouchard et al., "Aerobic Performance in Brothers, Dizygotic and Monozygotic Twins," *Medicine and Science in Sports and Exercise* 18 (1986): 639-646.



Bouchard and colleagues⁴ concluded that heredity accounts for between 25% and 50% of the variance in $\dot{V}O_{2\max}$ values. This means that of all factors influencing $\dot{V}O_{2\max}$, heredity alone is responsible for one-quarter to one-half of the total influence. World-class athletes who have stopped endurance training continue for many years to have high $\dot{V}O_{2\max}$ values in their sedentary, deconditioned state. Their $\dot{V}O_{2\max}$ values may decrease from 85 to 65 ml · kg⁻¹ · min⁻¹, but this deconditioned value is still very high compared with that of the general population.

Heredity also potentially explains the fact that some people have relatively high $\dot{V}O_{2\max}$ values yet have no history of endurance training. In a study that compared untrained men who had $\dot{V}O_{2\max}$ values below 49 ml · kg⁻¹ · min⁻¹ with untrained men who had $\dot{V}O_{2\max}$ values above 62.5 ml · kg⁻¹ · min⁻¹, those with high values were distinguished by having higher blood volumes, which contributed to higher stroke volumes and cardiac outputs at maximal intensities. The higher blood volumes in the high $\dot{V}O_{2\max}$ group were most likely genetically determined.¹¹

Thus, both genetic and environmental factors influence $\dot{V}O_{2\max}$ values. The genetic factors probably establish the boundaries for the athlete, but endurance training can push $\dot{V}O_{2\max}$ to the upper limit of these boundaries. Dr. Per-Olof Åstrand, one of the most highly recognized exercise physiologists during the second half of the 20th century, stated on numerous occasions that the best way to become a champion Olympic athlete is to be selective when choosing one's parents!

Sex

Healthy untrained girls and women have significantly lower $\dot{V}O_{2\max}$ values (20%-25% lower) than healthy untrained boys and men. Highly conditioned female endurance athletes have values much closer to those of highly trained male endurance athletes (i.e., only about 10% lower). This is discussed in greater detail in [chapter 19](#). Representative ranges of $\dot{V}O_{2\max}$ values for athletes and nonathletes are presented in [table 11.4](#) by age, sex, and sport.

High Responders and Low Responders

For years, researchers have found wide variations in the amount of improvement in $\dot{V}O_{2\max}$ with endurance training. Studies have demonstrated individual improvements in $\dot{V}O_{2\max}$ ranging from 0% to 50% or more, even in similarly fit subjects completing exactly the same training program.

TABLE 11.4 Maximal Oxygen Uptake Values ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) for Nonathletes and Athletes

Group or sport	Age group (years)	Males	Females
Nonathletes	10-19	47-56	38-46
	20-29	43-52	33-42
	30-39	39-48	30-38
	40-49	36-44	26-35
	50-59	34-41	24-33
	60-69	31-38	22-30
	70-79	28-35	20-27
Baseball and softball	18-32	48-56	52-57
Basketball	18-30	40-60	43-60
Bicycling	18-26	62-74	47-57
Canoeing	22-28	55-67	48-52
Football	20-36	42-60	
Gymnastics	18-22	52-58	36-50
Ice hockey	10-30	50-63	
Jockey	20-40	50-60	
Orienteering	20-60	47-53	46-60
Racquetball	20-35	55-62	50-60
Rowing	20-35	60-72	58-65
Skiing, alpine	18-30	57-68	50-55
Skiing, Nordic	20-28	65-94	60-75
Ski jumping	18-24	58-63	
Soccer	22-28	54-64	50-60
Speed skating	18-24	56-73	44-55
Swimming	10-25	50-70	40-60
Track and field, discus	22-30	42-55	*
Track and field, running	18-39	60-85	50-75
	40-75	40-60	35-60
Track and field, shot put	22-30	40-46	*
Volleyball	18-22		40-56
Weightlifting	20-30	38-52	*
Wrestling	20-30	52-65	

*Data not available.

In the past, exercise physiologists have assumed that these variations result from differing degrees of compliance with the training

program. People who comply with the program should, and do, have the highest percentage of improvement, and poor compliers should show little or no improvement. However, given the same training stimulus and full compliance with the program, substantial variations still occur in the percent improvement in $\dot{V}O_{2\max}$ for different people.

It is now evident that some of the response to a training program is also genetically determined. This is illustrated in [figure 11.16](#). Ten pairs of identical twins completed a 20-week endurance training program; the improvements in $\dot{V}O_{2\max}$, expressed as percentages, are plotted for each twin pair—twin A on the x-axis and twin B on the y-axis.²⁷ Notice the similarity in response of each twin pair. Yet across twin pairs, improvement in $\dot{V}O_{2\max}$ varied from 0% to 40%. These results, and those from other studies, indicate that there will be **high responders** (showing large improvement) and **low responders** (showing little or no improvement) among groups of people who participate in identical training programs. However, while genetic variants may be involved, such variants appear to be associated with the physiological mechanisms (increased cardiac output, expanded blood volume, improved muscle oxygen extraction) that underpin such differences.²⁰

Results from the HERITAGE Family Study also support a strong genetic component in the magnitude of increase in $\dot{V}O_{2\max}$ with endurance training. Families, including the biological mother and father and three or more of their children, trained 3 days a week for 20 weeks, initially exercising at a heart rate equal to 55% of their $\dot{V}O_{2\max}$ for 35 min per day and progressing to a heart rate equal to 75% of their $\dot{V}O_{2\max}$ for 50 min per day by the end of the 14th week, which they maintained for the last 6 weeks.³ The average increase in $\dot{V}O_{2\max}$ was about 17% but varied from 0% to more than 50%. [Figure 11.17](#) illustrates the improvement in $\dot{V}O_{2\max}$ for each subject in each family. Maximal heritability was estimated at 47%. Note that subjects who are high responders tend to be clustered in the same families, as are those who are low responders.

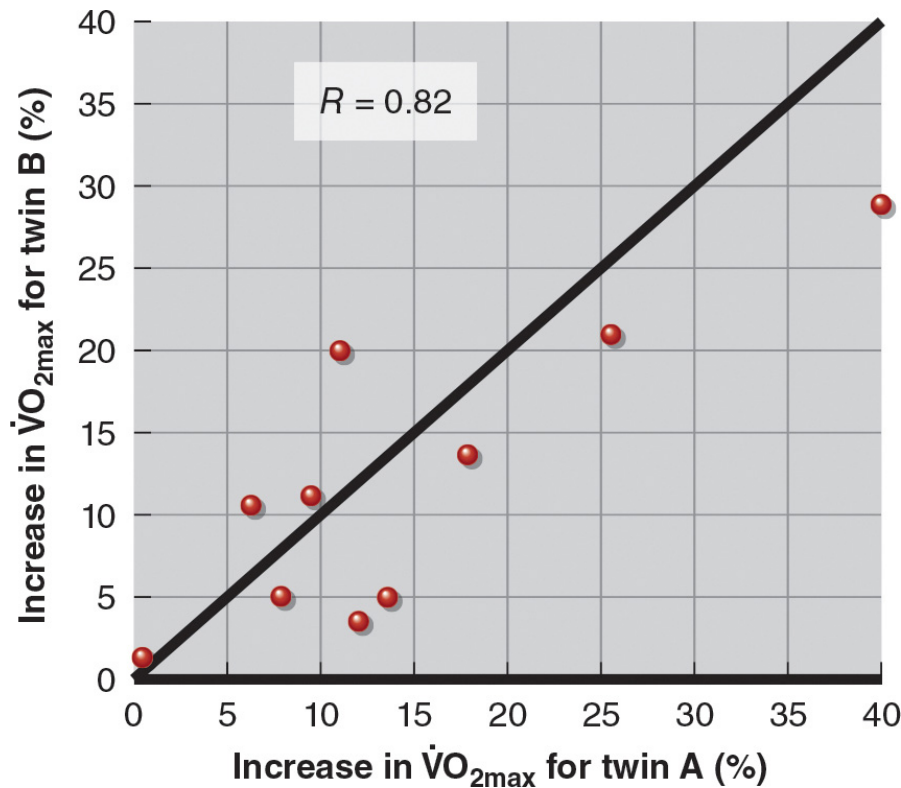


FIGURE 11.16 Variations in the percentage increase in $\dot{V}O_{2\max}$ for identical twins undergoing the same 20-week training program.

Reprinted by permission from D. Prud'homme et al., "Sensitivity of Maximal Aerobic Power to Training is Genotype-Dependent," *Medicine and Science in Sports and Exercise* 16, no. 5 (1984): 489-493.

It is clear that this is a genetic phenomenon, not a result of compliance or noncompliance. One must consider this important point when conducting training studies and designing training programs. Individual differences must always be accounted for.

Cardiorespiratory Endurance in Nonendurance Sports

Many people regard cardiorespiratory endurance as the most important component of physical fitness. Low endurance capacity leads to fatigue, even in activities that are not aerobic. For any athlete, regardless of the sport or activity, fatigue represents a major deterrent to optimal performance. Even minor fatigue can hinder the athlete's total performance:

- Muscular strength is decreased.
- Reaction and movement times are prolonged.
- Agility and neuromuscular coordination are reduced.

- Whole-body movement speed is slowed.
- Concentration and alertness are reduced.

The decline in concentration and alertness associated with fatigue is particularly important. The athlete can become careless and more prone to serious injury, especially in contact sports. Even though these decrements in performance might be small, they can be just enough to cause an athlete to miss the critical free throw in basketball, the strike zone in baseball, or the 20 ft (6 m) putt in golf.

All athletes can benefit from improving their cardiorespiratory endurance. Even golfers, whose sport demands little in the way of aerobic endurance, can benefit. Improved endurance can allow golfers to complete a round of golf with less fatigue and to better withstand long periods of walking and standing.

For the sedentary, middle-aged adult, numerous health factors indicate that cardiovascular endurance should be the primary emphasis of training. Training for health and fitness is discussed at length in [part VII](#) of this book.

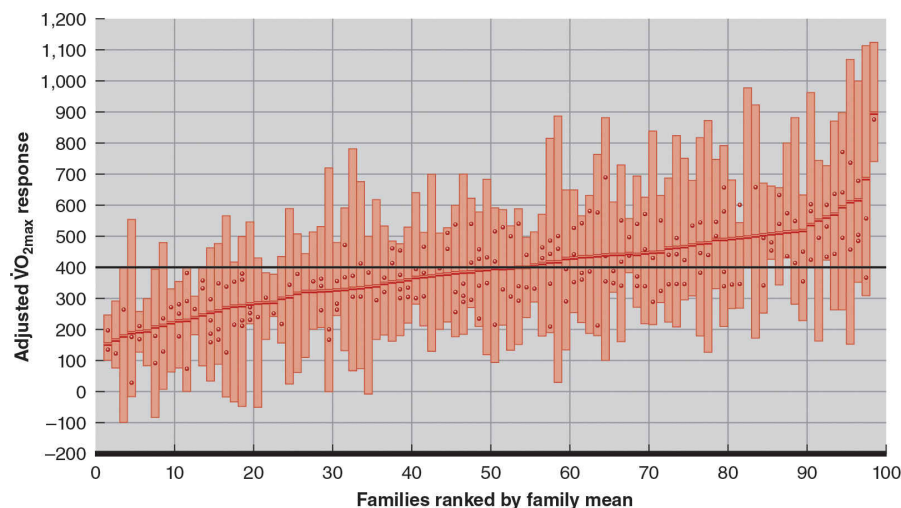


FIGURE 11.17 Variations in the improvement in $\dot{V}O_{2\max}$ following 20 weeks of endurance training by families. Values represent the changes in $\dot{V}O_{2\max}$ in ml/min, with an average increase of 393 ml/min. Data for each family are enclosed within a bar, and each family member's value is represented as a dot within the bar.

Adapted by permission from C. Bouchard, P. An, T. Rice, J.S. Skinner, J.H. Wilmore et al., "Familial Aggregation of $\dot{V}O_{2\max}$ Response to Exercise Training. Results from HERITAGE Family Study," *Journal of Applied Physiology* 87 (1999): 1003-1008.

The extent of endurance training needed varies considerably from one sport to the next and from one athlete to the next. It depends on the athlete's current endurance capacity and the endurance demands of the chosen activity. However, adequate cardiovascular conditioning must be the foundation of any athlete's general conditioning program.

In Review

- Although improvements in $\dot{V}O_{2\max}$ eventually plateau, endurance performance can continue to improve for years with continued training.
- An individual's genetic makeup predetermines a range for that person's $\dot{V}O_{2\max}$ and accounts for 25% to 50% of the variance in $\dot{V}O_{2\max}$ values.
- Heredity also largely explains individual variations in response to identical training programs.
- Highly conditioned female endurance athletes have $\dot{V}O_{2\max}$ values only about 10% lower than those of highly conditioned male endurance athletes.
- All athletes, regardless of their sport or event, can benefit from maximizing their cardiorespiratory endurance.

Aerobic Deconditioning

Issues related to deconditioning are particularly relevant to bed rest associated with diseases and disability as well as to the space program, since weightlessness and bed rest cause similar declines in $\dot{V}O_{2\max}$. According to a recent analysis, 80 studies with a total of 949 participants have been published since 1949 that reported the effects of total bed rest on $\dot{V}O_{2\max}$.²⁹ The studies were conducted mainly in young (age range 22-34 years), male (>90%) subjects with bed rest lasting from 1 to 90 days. Declines in $\dot{V}O_{2\max}$ were fairly linear throughout periods of prolonged bed rest.

Surprisingly, while body weight and lean body mass both decline in response to bed rest, those changes were unrelated to the decline in $\dot{V}O_{2\max}$. The most important predictor of how much $\dot{V}O_{2\max}$ dropped was the subjects' fitness level at the beginning of the bed rest period. Higher initial $\dot{V}O_{2\max}$ levels were associated with larger declines in $\dot{V}O_{2\max}$.

Adaptations to Anaerobic Training

In muscular activities that require near-maximal force production for relatively short periods of time, such as sprinting, much of the energy needs are met by the ATP-phosphocreatine (PCr) system and the anaerobic breakdown of muscle glycogen (glycolysis). The following sections focus on the trainability of these two systems.

Changes in Anaerobic Power and Anaerobic Capacity

Exercise scientists have had difficulty agreeing on an appropriate laboratory or field test to measure anaerobic power. Unlike the situation with aerobic power, for which $\dot{V}O_{2\max}$ is generally agreed to be the gold standard measurement, no single test adequately measures anaerobic power. Most research has been conducted through use of three different tests of either anaerobic power, anaerobic capacity, or both: the Wingate anaerobic test, the critical power test, and the maximal accumulated oxygen deficit test. Of these three, the Wingate test has been the most widely used. Despite the limitations inherent in each of these methods, they remain our only indirect indicators of the metabolic potential of anaerobic capacity.



As described in [chapter 9](#), the Wingate anaerobic test is commonly used to measure anaerobic power. Peak power output, the highest mechanical power achieved during the first 5 to 10 s, is considered an index of anaerobic power. The mean power output is computed as the average power output over the total 30 s period, and one obtains total work simply by multiplying the mean power output by 30 s. Mean power output and total work have both been used as indexes of anaerobic capacity.

With anaerobic training, such as sprint training on the track or on a cycle ergometer, there are increases in both peak anaerobic power and anaerobic capacity. However, results have varied widely across

studies, from those that showed only minimal increases to those showing increases of up to 25%.

Adaptations in Muscle with Anaerobic Training

With anaerobic training, which includes sprint training and resistance training, there are changes in skeletal muscle that specifically reflect muscle fiber recruitment for these types of activities. As discussed in [chapter 1](#), at higher intensities, type II muscle fibers are recruited to a greater extent, but not exclusively, because type I fibers continue to be recruited. Overall, sprint and resistance activities use the type II muscle fibers significantly more than do aerobic activities. Consequently, both type IIa and type IIx muscle fibers undergo an increase in their cross-sectional areas. The cross-sectional area of type I fibers also is increased but usually to a lesser extent. Furthermore, with sprint training there appears to be a reduction in the percentage of type I fibers and an increase in the percentage of type II fibers, with the greatest change in type IIa fibers. In two of these studies, in which subjects performed 15 to 30 s all-out sprints, the type I percentage decreased from 57% to 48% and type IIa increased from 32% to 38%.^{17,18} This shift of type I to type II fibers is not typically seen with resistance training.

Adaptations in the Energy Systems

Just as aerobic training produces changes in the aerobic energy system, anaerobic training alters the ATP-PCr and anaerobic glycolytic energy systems. These changes are not as obvious or predictable as those that result from endurance training, but they do improve performance in anaerobic activities.

Adaptations in the ATP-PCr System

Activities that emphasize maximal muscle force production, such as sprinting and weightlifting events, rely most heavily on the ATP-PCr system for energy. Maximal effort lasting less than about 6 s places the greatest demands on the breakdown and resynthesis of ATP and PCr. Costill and coworkers reported their findings from a study of resistance training and its effects on the ATP-PCr system.⁸ Their participants trained by performing maximal knee extensions. One leg was trained using 6 s maximal work bouts that were repeated 10

times. This type of training preferentially stressed the ATP-PCr energy system. The other leg was trained with repeated 30 s maximal bouts, which instead preferentially stressed the glycolytic system.

The two forms of training produced the same muscular strength gains (about 14%) and the same resistance to fatigue. As seen in [figure 11.18](#), the activities of the anaerobic muscle enzymes creatine kinase and myokinase increased as a result of the 30 s training bouts but were almost unchanged in the leg trained with repeated 6 s maximal efforts. This finding leads us to conclude that maximal sprint bouts (6 s) might improve muscular strength but contribute little to the mechanisms responsible for ATP and PCr breakdown. Data have been published, however, that show improvements in ATP-PCr enzyme activities with training bouts lasting only 5 s.

Regardless of the conflicting results, these studies suggest that the major value of training bouts that last only a few seconds (sprints) is the development of muscular strength. Such strength gains enable the individual to perform a given task with less effort, which reduces the risk of fatigue. Whether these changes allow the muscle to perform more anaerobic work remains unanswered, although a 60 s sprint-fatigue test suggests that short sprint-type anaerobic training does not enhance anaerobic endurance.⁸

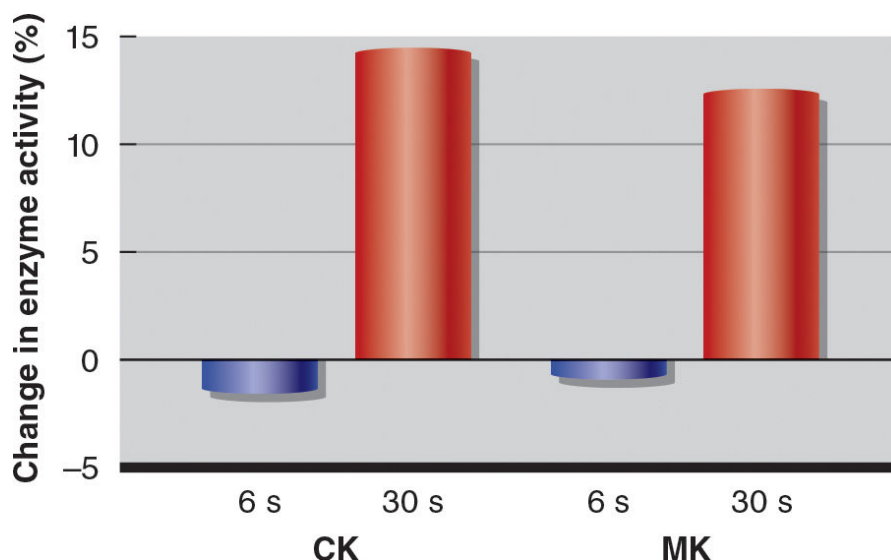


FIGURE 11.18 Changes in creatine kinase (CK) and muscle myokinase (MK) activities as a result of 6 s and 30 s bouts of maximal anaerobic training.

Adaptations in the Glycolytic System

Anaerobic training (30 s bouts) increases the activities of several key glycolytic enzymes. The most frequently studied glycolytic enzymes are phosphorylase, phosphofructokinase (PFK), and lactate dehydrogenase (LDH). The activities of these three enzymes increased 10% to 25% with repeated 30 s training bouts but changed little with short (6 s) bouts that stress primarily the ATP-PCr system.⁸ In another study, 30 s maximal all-out sprints significantly increased hexokinase (56%) and PFK (49%) but not total phosphorylase activity or LDH.²¹

Because both PFK and phosphorylase are essential to the anaerobic yield of ATP, such training might enhance glycolytic capacity and allow the muscle to develop greater tension for a longer period of time. However, as seen in [figure 11.19](#), this conclusion is not supported by results of a 60 s sprint performance test, in which the subjects performed maximal knee extension and flexion. Power output and the rate of fatigue (shown by a decrease in power production) were affected to the same degree after sprint training with either 6 s or 30 s training bouts. Thus, we must conclude that performance gains with these forms of training result from improvements in strength rather than improvements in the anaerobic yield of ATP.

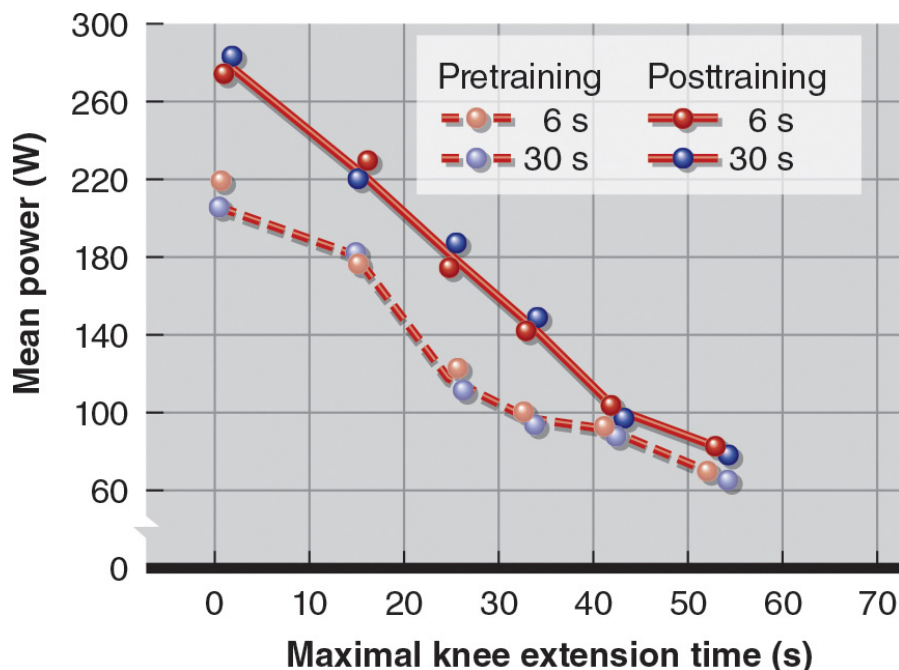


FIGURE 11.19 Performance in a 60 s sprint bout after training with 6 s and 30 s anaerobic bouts. Subjects are the same as in [figure 11.18](#).

Adaptations to High-Intensity Interval Training

In [chapter 9](#) we introduced a special form of training using short bursts of very intense cycling, interspersed with up to a few minutes of rest or low-intensity cycling for recovery.¹⁴ High-intensity interval training (HIIT) is a time-efficient way to induce many aerobic training benefits normally associated with continuous running, cycling, or swimming.

Adaptations to HIIT mirror those associated with more traditional aerobic training. In one study, untrained young subjects performed four to six 30 s sprints separated by 4 min of recovery, three times a week. These men showed the same beneficial changes in their heart, blood vessels, and muscles as another group who underwent a traditional training program involving up to an hour of continuous cycling, 5 days per week. Improvements in exercise performance—whether measured as cycling time to exhaustion at a fixed work intensity or in time trials that more closely resemble normal athletic competition—were comparable between groups, despite considerable differences in training time commitment.¹⁴ High-intensity interval training appears to stimulate some of the same molecular signaling pathways that regulate skeletal muscle remodeling in response to endurance training, including mitochondrial biogenesis and changes in the capacity for carbohydrate and fat transport and oxidation.

In Review

- Anaerobic training bouts improve both anaerobic power and anaerobic capacity.
- The performance improvement noted with sprint-type anaerobic training appears to result more from strength gains than from improvements in the functioning of the anaerobic energy systems.
- Anaerobic training increases the ATP-PCr and glycolytic enzymes but has no effect on the oxidative enzymes.
- Conversely, aerobic training increases the oxidative enzymes but has little effect on the ATP-PCr or glycolytic enzymes.

- Adaptations to HIIT mirror those associated with more traditional aerobic training. High-intensity interval training appears to stimulate some of the same molecular signaling pathways that regulate skeletal muscle remodeling in response to endurance training, including mitochondrial biogenesis and changes in the capacity for carbohydrate and fat transport and oxidation.
-

RESEARCH PERSPECTIVE 11.2

Brief, Intense Stair Climbing

Low cardiorespiratory fitness is a strong predictor of cardiovascular disease and death. Public health guidelines generally recommend 150 min/week of moderate-intensity physical activity to achieve health benefits, but less than 15% of North Americans meet that recommendation. Lack of time and lack of necessary equipment are the two most commonly cited reasons for not achieving the recommended daily physical activity. Because of this, public health researchers and exercise physiologists are interested in finding easily accessible and briefer exercise protocols that achieve the same health benefits as the current 150 min/week of moderate-intensity exercise recommendation.

High-intensity interval training, or HIIT, which involves brief intermittent bursts of high-intensity exercise separated by recovery periods, improves cardiorespiratory fitness. Sprint interval training has been shown to improve fitness and insulin sensitivity to the same extent as a moderate-intensity continuous exercise protocol that required five times as much time to complete. Knowing this, a research team at McMaster University in Canada recently conducted a series of studies to see if brief, intense stair climbing, a readily available high-intensity activity, could improve cardiorespiratory fitness.¹ In these studies, young, sedentary women performed an acute exercise comparison to ensure that the stair-climbing protocol elicited the same acute physiological responses as a classical sprint interval training bout on a stationary bike, and a 6-week training intervention. In the intervention period, subjects were instructed to perform three 20 s bursts of all-out intensity going up the stairs as fast as possible with 2 min of recovery between bouts, 3 days/week.

Work output, heart rate, blood lactate, and RPE responses were the same between the brief intense stair-climbing protocol and the classical sprint interval training protocol. Following 6 weeks of stair-climbing training, cardiorespiratory fitness ($\dot{V}O_{2peak}$) had improved by 12%, which is similar to other studies using a cycle ergometer to administer sprint interval training. Importantly, the study team also reported that the participants completed 99% of all the training sessions and that the average time required to complete the training was ≤ 9 min/week. Overall, brief, intense stair climbing is a time-

efficient and easily accessible mode of exercise that can increase cardiorespiratory fitness in sedentary adults.

RESEARCH PERSPECTIVE 11.3

Do Ice Baths Increase Recovery and Endurance Performance?

Postexercise cold-water immersion has become increasingly popular in athletic training programs because of the belief that it speeds up recovery. However, few studies have scientifically investigated the effect of postexercise cold-water immersion therapy on the adaptive responses to endurance training, and no studies have examined these effects with sprint interval training. Of the research studies that have been conducted, the results have been conflicting. Some suggest that cold-water immersion may stimulate muscle mitochondria biogenesis and allow for better recovery and harder training in subsequent bouts of exercise, while other say that postexercise cold-water immersion may counteract the molecular processes related to vascular remodeling and have detrimental long-term effects on skeletal muscle adaptations to endurance training. Overall, the utility of postexercise cold-water immersion to enhance recovery and subsequent performance is still unclear.

A recent study conducted at Victoria University in Australia investigated the effects of cold-water immersion on mitochondrial content and function within the muscle (1) following a single bout of sprint interval (HIIT) exercise and (2) after a 6-week HIIT intervention.⁷ The researchers recruited healthy, recreationally active men and split them into two groups. One group received postexercise cold-water immersion after each training bout, while the other group performed the same training but had a passive recovery without cold water. After a baseline familiarization trial, subjects performed a single bout of HIIT training followed by a skeletal muscle biopsy. After the 6-week training intervention, another muscle biopsy was done along with posttraining time-trial and $\dot{V}O_{2\max}$ testing. The investigators analyzed the skeletal muscle biopsies for p-AMPK, p-p38 MAPK, p-p53, and PGC-1 α , which are markers of mitochondrial content and function. In short, the investigators did not find any effect of cold-water immersion on any of their measurements. There were no differences between the group of participants who were treated with cold-water immersion after each exercise bout and the control group, who performed the exercise training without cold-water immersion during recovery. This program of HIIT increased $\dot{V}O_{2\max}$ and time trial performance without any effect on markers of mitochondrial content or function.

While these findings suggest that cold-water immersion is not detrimental to endurance adaptations after sprint interval training, they also suggest that

cold-water immersion does not provide any benefit to endurance training adaptations or improvements in fitness and performance.

As discussed in [chapter 9](#), athletes who already train vigorously can likewise improve performance by integrating HIIT into their training regimens. However, the mechanisms for these improvements appear to differ.¹³ The rapid increases in skeletal muscle oxidative enzymes seen in previously untrained exercisers are not apparent in already trained individuals who add HIIT to their workouts. The underlying adaptations for improved performance in these athletes are not well understood.

Specificity of Training and Cross-Training

Physiological adaptations in response to physical training are highly specific to the nature of the training activity. Furthermore, the more specific the training program is to a given sport or activity, the greater the improvement in performance in that sport or activity. As discussed in [chapter 9](#), the concept of **specificity of training** is very important for all physiological adaptations.

This concept is also important in testing of athletes. As an example, to accurately measure endurance improvements, athletes should be tested while engaged in an activity similar to the sport or activity in which they usually participate. Consider one study of highly trained rowers, cyclists, and cross-country skiers. Their $\dot{V}O_{2\max}$ was measured while they performed two types of work: uphill running on a treadmill and maximal performance of their sport-specific activity.³³ A key finding, shown in [figure 11.20](#), was that the $\dot{V}O_{2\max}$ attained by all the athletes during their sport-specific activity was as high as or higher than the values obtained on the treadmill. For many of these athletes, $\dot{V}O_{2\max}$ was substantially higher during their sport-specific activity.

A highly creative design for studying the concept of specificity of training involves one-legged exercise training, with the untrained opposite leg used as the control. In one study, subjects were placed into three groups: a group that sprint trained one leg and endurance trained the other, a group that sprint trained one leg and did not train the other, and a group that endurance trained one leg and did not

train the other.³⁰ Improvement in $\dot{V}O_{2\max}$ and lowered heart rate and blood lactate response at submaximal work rates were found only when exercise was performed with the endurance-trained leg.

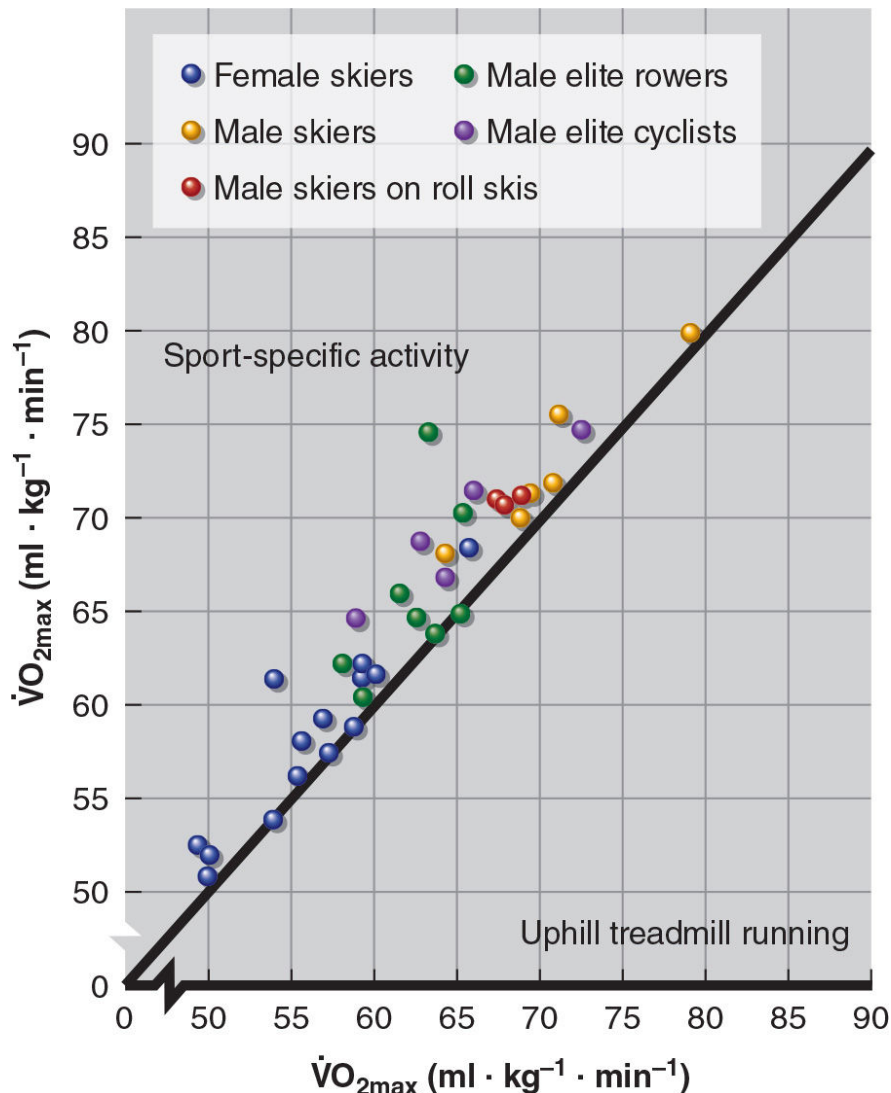


FIGURE 11.20 $\dot{V}O_{2\max}$ values measured during uphill treadmill running versus sport-specific activities in selected groups of athletes.

Adapted by permission from S.B. Strømme, F. Ingjer, and H.D. Meen, "Assessment of Maximal Aerobic Power in Specifically Trained Athletes," *Journal of Applied Physiology* 42 (1977): 833-837.

Much of the training response occurs in the specific muscles that have been trained, possibly even in individual motor units in a specific muscle. This observation applies to metabolic as well as cardiorespiratory responses to training. [Table 11.5](#) shows the activities of selected muscle enzymes from the three energy systems

for untrained, anaerobically trained, and aerobically trained men. The table shows that aerobically trained muscles have significantly lower glycolytic enzyme activities. Thus, they might have less capacity for anaerobic metabolism or might rely less on energy from glycolysis. More research is needed to explain the implications of the muscular changes accompanying both anaerobic and aerobic training, but this table clearly illustrates the high degree of specificity to a given training stimulus.

Cross-training refers to training for more than one sport at the same time or training several different fitness components (such as endurance, strength, and flexibility) at one time. The athlete who trains by swimming, running, and cycling in preparation for competing in a triathlon is an example of the former, and the athlete involved in heavy resistance training and high-intensity cardiorespiratory training at the same time is an example of the latter.

TABLE 11.5 Selected Muscle Enzyme Activities (mmol · g⁻¹ · min⁻¹) for Untrained, Anaerobically Trained, and Aerobically Trained Men

	Untrained	Anaerobically trained	Aerobically trained
Aerobic enzymes			
Oxidative system			
Succinate dehydrogenase	8.1	8.0	20.8a
Malate dehydrogenase	45.5	46.0	65.5a
Carnitine palmityl transferase	1.5	1.5	2.3a
Anaerobic enzymes			
ATP-PCr system			
Creatine kinase	609.0	702.0 ^a	589.0
Myokinase	309.0	350.0 ^a	297.0
Glycolytic system			
Phosphorylase	5.3	5.8	3.7 ^a
Phosphofructokinase	19.9	29.2 ^a	18.9
Lactate dehydrogenase	766.0	811.0	621.0

^aSignificant difference from the untrained value.

RESEARCH PERSPECTIVE 11.4

Age and Responses to HIIT

Maximal oxygen consumption ($\dot{V}O_{2max}$) is one of the strongest predictors of cardiovascular health span and mortality. Even with regular aerobic activity, $\dot{V}O_{2max}$ declines ~1% per year with age, and this decline accelerates in older age. Consequently, older adults, who already have a higher risk for cardiovascular disease and mortality, could benefit the most from interventions

that increase $\dot{V}O_{2\max}$. High-intensity interval training (HIIT) yields effective improvements in aerobic fitness and cardiovascular health in healthy young and middle-aged adults. Because of the relatively short time commitment and significant improvements in fitness achieved, HIIT may be an especially valuable strategy for improving $\dot{V}O_{2\max}$ in older adults. However, few research studies have examined how age affects the aerobic training response to HIIT in older adults.

A recent study of 94 healthy men and women ranging from 20 to 83 years of age sought to determine how age affected improvements in $\dot{V}O_{2\max}$ following HIIT training.³² In this study, participants with similar pretest $\dot{V}O_{2\max}$ values relative to age were tested immediately before and immediately following an 8-week HIIT intervention. During the intervention, the study participants completed supervised HIIT training with a targeted intensity of 90% to 95% of maximal heart rate, three times a week. After the HIIT intervention, all of the subjects improved their $\dot{V}O_{2\max}$. In order to examine age differences, the subjects were separated into six age groups (20-29 years, 30-39 years, 40-49 years, 50-59 years, 60-69 years, and 70+ years). All of the groups improved their $\dot{V}O_{2\max}$ with no differences between age groups. In contrast to age, the percentage improvements in $\dot{V}O_{2\max}$ were predicted by their baseline $\dot{V}O_{2\max}$, with the least fit people showing the greatest improvements regardless of their age group. Healthy individuals of average fitness can improve $\dot{V}O_{2\max}$ through HIIT, regardless of their age, and HIIT may be a useful strategy to improve fitness and slow the decline in $\dot{V}O_{2\max}$ in healthy aging.

For the athlete training for cardiorespiratory endurance and strength at the same time, the studies conducted to date indicate that gains in strength, power, and endurance can result. However, the gains in muscular strength and power are less when strength training is combined with endurance training than when strength training alone is done. The opposite does not appear to be true: Improvement of aerobic power with endurance training does not appear to be attenuated by inclusion of a resistance training program. In fact, short-term endurance can be increased with resistance training. Although earlier studies supported the conclusion that concurrent strength and endurance training limits gains in strength and power, McCarthy and colleagues²³ reported similar gains in strength, muscle hypertrophy, and neural activation in a group of previously untrained subjects who underwent concurrent high-intensity strength training

and cycle endurance training compared with a group who performed only high-intensity strength training.

In Review

- For athletes to maximize cardiorespiratory gains from training, the training should be specific to the type of activity that an athlete usually performs.
 - The program must be carefully matched with the athlete's individual needs to maximize the physiological adaptations to training, thereby optimizing performance.
 - Resistance training in combination with endurance training does not appear to restrict improvement in aerobic power and may increase short-term endurance, but it can limit improvement in strength and power when compared with gains from resistance training alone.
-

IN CLOSING

In this chapter, we examined how the cardiovascular, respiratory, and metabolic systems adapt to aerobic and anaerobic training, as well as HIIT training. The focus was on how these adaptations can improve both aerobic and anaerobic performance. This chapter concludes our review of how body systems respond to both acute and chronic exercise. Now that we have completed our examination of how the body responds to internal challenges induced by various types, durations, and intensities of exercise, we turn our attention to the external environment. In the next part of the book, we focus on the body's adaptations to varying environmental conditions, beginning in the next chapter by considering how external temperature can affect performance.

KEY TERMS

aerobic training
anaerobic training
athlete's heart
capillary-to-fiber ratio
cardiac hypertrophy
cross-training
Fick equation
glycogen sparing
high responders
low responders
mitochondrial oxidative enzymes
oxygen transport system
specificity of training
submaximal endurance

STUDY QUESTIONS

1. Differentiate between muscular endurance and cardiovascular endurance.
2. What is maximal oxygen uptake ($\dot{V}O_{2\max}$)? How is it defined physiologically, and what determines its limits?
3. Of what importance is $\dot{V}O_{2\max}$ to endurance performance? Why does the competitor with the highest $\dot{V}O_{2\max}$ not always win?
4. Describe the changes in the oxygen transport system that occur with endurance training.
5. What is possibly the most important adaptation that the body makes in response to endurance training, which allows for an increase in both $\dot{V}$

$\dot{V}O_{2\max}$ and performance? Through what mechanisms do these changes occur?

6. What are the theoretical reasons given for the resting bradycardia that accompanies endurance exercise training?
7. What metabolic adaptations occur in response to endurance training?
8. Explain the two theories that have been proposed to account for limitations to aerobic performance that may be altered by endurance training. Which of these has the greatest validity today?
9. What is the most important predictor of how much $\dot{V}O_{2\max}$ will decline with inactivity or bed rest?
10. How important is genetic potential in a developing young athlete?
11. What adaptations have been shown to occur in muscle fibers with anaerobic training?
12. Discuss specificity of anaerobic training with respect to enzyme changes in muscle.
13. Can athletes who already train vigorously still improve performance by integrating **HIIT** into their training regimens? In what way are adaptive mechanisms different from those seen in untrained individuals who undergo HIIT?
14. Why is cross-training beneficial to endurance athletes? How does it benefit sprint and power athletes?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.